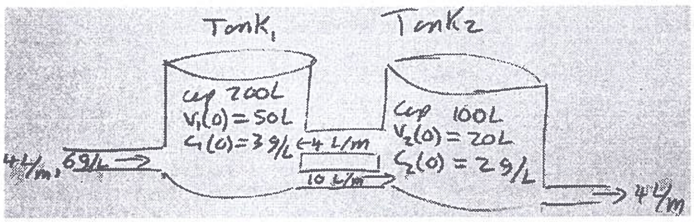
$V_1 = 50 - 2t$
$V_2 = 20 + 2t$

Name

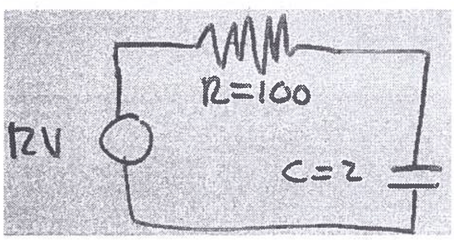
1. Show appropriate work.

1.1. Complete the ODE model $y' = A\,y + f$ and $y(0) = y_0$ for $y = \begin{pmatrix} s_1 \\ s_2 \end{pmatrix}$ with s_i the salt in tank i.



$A = \begin{pmatrix} -10/V_1 & 4/V_2 \\ 10/V_1 & -8/V_2 \end{pmatrix}$ $f = \begin{pmatrix} (4)(6) \\ 0 \end{pmatrix}$ and $y_0 = \begin{pmatrix} 3(50) \\ 2(20) \end{pmatrix}$

1.2. Write down the ODE model for the charge $q(t)$ on the capacitor.



ODE = $12 + q'100 + q/2 = 0$

1.3. A thousand immortal unicorns (50:50 m/f) land on Atlantis. Unicorn births are twins 3 times a year. Write down and solve a model for $f(t)$ the number of females.

f(t) = $500e^{3t}$

ODE $f' = \frac{1}{2}(2)(3)f = 3f$

$f(0) = 500$

2. Solve $2\,y' = 6\,y + e^{-7\,t}$. Show appropriate work.

y(t) = $-\frac{1}{20}e^{-7t} + ce^{3t}$

$y' - 3y = \frac{1}{2}e^{-7t}$

$v(t) = e^{-3t}$

$(e^{-3t}y)' = \frac{1}{2}e^{-3t}e^{-7t}$

$= \frac{1}{2}e^{-10t}dt$

$e^{-3t}y = \frac{1}{2}\int e^{-10t}dt + c$

$e^{-3t}y = \frac{1}{2}\frac{e^{-10t}}{-10} + c$

$y = -\frac{1}{20}e^{-10t}e^{3t} + ce^{3t}$

3. Solve $\left(-2\,t\sin\left(t^2\right) + y^2\right)\mathrm{dt} + (1 + 2\,t\,y)\,\mathrm{dy} = 0$. Show appropriate work.

$\cos(t^2) + y + ty^2 = C$

Exact

$\partial/\partial t\,(1+2ty) = 2y$

$\partial/\partial y\,(-2t\sin(t^2) + y^2) = 2y$

$S = \int 1 + 2ty\,dy = y + ty^2 + h(t)$

$\partial S/\partial t = 0 + y^2 + h'(t) = -2t\sin(t^2) + y^2$

$h'(t) = -2t\sin(t^2)$

$h(t) = \int -2t\sin(t^2)dt = -\int \sin(u)du = \cos(u)$

$u = t^2$

$du = 2t\,dt$

4. Solve $t\,y' - 3\,y = t^5$ with $y(1) = 0$ Show appropriate work.

$\boxed{\mathbf{y(t) =}\ t^5/2 + \tfrac{1}{2}t^3}$

$y' - 3/t\, y = t^4$

$v(t) = e^{-3\ln(t)} = e^{\ln(t^{-3})} = t^{-3}$

$(t^{-3}y)' = t^4 t^{-3} = t$

$t^{-3}y = \int t\,dt + c = t^2/2 + c$

$y = t^5/2 + c\,t^3$

$1 = 1/2 + c \Rightarrow c = 1/2$

5. Solve $y\,y' = 2\,e^{2\,t+y^2}$ with $y(1) = 0$. Show appropriate work.

$$\boxed{-\tfrac{1}{2}e^{-y^2} = e^{2t} + (-e^{2} - \tfrac{1}{2})}$$

$$y\,dy/dt = 2e^{2t} \cdot e^{y^2}$$

$$\int y e^{-y^2}\,dy = \int 2e^{2t}\,dt + C$$

$$= e^{2t} + C$$

$$\boxed{\begin{aligned} u &= y^2 \\ du &= 2y\,dy \end{aligned}}$$

$$= e^{2t} + C$$

$$\tfrac{1}{2}\int e^{-u}\,du = e^{2t} + C$$

$$-\tfrac{1}{2}e^{-y^2} = e^{2} + C$$

$$-\tfrac{1}{2}e^{-0} = e^{2} + C$$

$$-\tfrac{1}{2} - e^{2}$$

$$c = -\tfrac{1}{2} - e^{2}$$